📘 Today's Lesson: DTP & VTP (11.1.2026)

CCNA note: DTP and VTP were removed from the 200-301 exam topics list, but you may still get basic questions about their purpose/function.

🔌 Part 1: DTP – Dynamic Trunking Protocol

What it is: Cisco proprietary Layer 2 protocol that lets Cisco switches dynamically negotiate whether an interface becomes an **access port** or **trunk port** without manual configuration.

- Enabled by default on Cisco switch interfaces.
- Only works between Cisco devices.
- Recommended for security: **disable DTP** and manually configure access/trunk ports.
- DTP will **not** form a trunk with a router/PC. For router-on-a-stick, manually configure the switch port as a trunk.
- Access mode disables DTP negotiation.
- Manual trunk mode **still sends DTP frames** unless you also use `switchport nonegotiate`.

🧭 DTP Administrative Modes

Mode	Behavior
`switchport mode access`	Static access; DTP disabled
`switchport mode trunk`	Manual trunk; still sends DTP unless `nonegotiate`
`switchport mode dynamic desirable`	Actively tries to form a trunk
`switchport mode dynamic auto`	Passive; forms trunk only if the other side actively tries

- Older switches default: **dynamic desirable**
- Newer switches default: **dynamic auto**
- Both new switches default to auto + auto → **access**, not trunk.

📊 DTP Operational Result Matrix

Local \ Peer	Access	Dynamic Auto	Dynamic Desirable	Trunk
Access	Access	Access	Access	⚠️ Misconfig
Dynamic Auto	Access	Access	Trunk	Trunk
Dynamic Desirable	Access	Trunk	Trunk	Trunk
Trunk	⚠️ Misconfig	Trunk	Trunk	Trunk

⚠️ Access + Trunk / Trunk + Access = misconfiguration; traffic will not pass correctly. Do not use in production.

🏷️ Trunk Encapsulation Negotiation

- Switches supporting both **dot1q** and **ISL** can negotiate encapsulation via DTP.
- Default: `switchport trunk encapsulation negotiate`
- To manually configure a trunk on a switch supporting both, you must set encapsulation to `dot1q` or `isl` first; you cannot leave it as `negotiate`.
- If both support ISL, **ISL is preferred** over dot1q.
- DTP frames are sent in **VLAN 1** for ISL, or the **native VLAN** for dot1q; default native VLAN is VLAN 1.
- `show interfaces switchport` field **Negotiation of Trunking** is:
  - **On** for dynamic desirable, dynamic auto, or trunk mode.
  - **Off** for access mode or when `switchport nonegotiate` is used.

🧪 Part 2: VTP – VLAN Trunking Protocol

What it is: Cisco proprietary protocol that lets you configure VLANs on a central **VTP server**; **VTP clients** synchronize their VLAN database to the server.

- Designed for large networks with many VLANs.
- Rarely used today; generally recommended **not** to use it.
- VTP only syncs the **VLAN database** – it does **not** assign interfaces to VLANs. You still configure `switchport access vlan X` on each switch.
- VTP advertisements are sent only on **trunk ports**.

🔁 VTP Revision Number

- Revision number increases when a VLAN is added, modified, or deleted.
- Highest revision number = newest VLAN database.
- VTP servers also act as clients: they will sync to another server with a higher revision number.
- ⚠️ Danger: connecting an old switch with a higher revision number and matching domain can overwrite the network VLAN database, causing outages.
- Reset revision number to 0 by:
  1. Changing VTP domain to an unused name.
  2. Changing VTP mode to transparent.
- `vtp reset` is **not** a valid method.

🧩 VTP Modes

Mode	Can add/modify/delete VLANs?	Stores VLAN DB in NVRAM?	Syncs to server?	Advertises/forwards?
Server	Yes	Yes	Yes, if higher revision received	Yes
Client	No	No, except VTPv3	Yes	Forwards advertisements
Transparent	Yes, local only	Yes	No	Forwards same-domain advertisements; does not advertise own DB

- Default Cisco switch mode: **VTP server**.
- Default domain: **NULL**.
- If a switch with NULL/no domain receives an advertisement with a domain name, it automatically joins that domain.
- Default existing VLANs: 5 → VLAN 1 and 1002–1005.

🗒️ VTP Versions

- **VTPv1**: default.
- **VTPv2**: adds Token Ring VLAN support; otherwise little difference.
- **VTPv3**: supports extended VLAN range 1006–4094; VTPv3 clients store VLAN DB in NVRAM.
- VTPv1/v2 support only VLANs up to 1005 locally.
- Changing VTP version increases revision number.

🛠️ Commands Used & Modes

DTP Commands

Command	Mode	Purpose
`switchport mode access`	Interface config	Static access; disables DTP
`switchport mode trunk`	Interface config	Manual trunk; still sends DTP
`switchport mode dynamic desirable`	Interface config	Actively negotiate trunk
`switchport mode dynamic auto`	Interface config	Passively accept trunk
`switchport nonegotiate`	Interface config	Stop sending DTP frames
`switchport trunk encapsulation negotiate`	Interface config	Default encapsulation negotiation
`switchport trunk encapsulation dot1q`	Interface config	Set dot1q encapsulation
`switchport trunk encapsulation isl`	Interface config	Set ISL encapsulation
`show interfaces G0/0 switchport`	EXEC / privileged EXEC	Show admin/operational mode, DTP negotiation, encapsulation
`show interfaces switchport`	EXEC / privileged EXEC	Same for all interfaces

VTP Commands

Command	Mode	Purpose
`vtp domain CISCO`	Global config	Set VTP domain name
`vtp mode server`	Global config	Set VTP server mode
`vtp mode client`	Global config	Set VTP client mode
`vtp mode transparent`	Global config	Set VTP transparent mode
`vtp version 2`	Global config	Change VTP version; increases revision
`vlan 10`	Global config	Create VLAN; enters VLAN config
`name engineering`	VLAN config	Name the VLAN

Command	Mode	Purpose
`show vtp status`	EXEC / privileged EXEC	Show domain, mode, version, revision, VLAN counts
`show vlan brief`	EXEC / privileged EXEC	Show VLAN database

❓ End-of-Lesson Quiz

Q1. SW1 and SW2 are connected, both new switches, interfaces operating as access ports. SW2's power supply fails, so you replace it with an old spare switch. You reset the old switch's configuration before connecting it, but a trunk forms between SW1 and the old switch. What could be the cause?

- A. Interfaces on old switches default to `switchport mode trunk`
- B. Interfaces on old switches default to `switchport mode dynamic desirable`
- C. Access ports are a feature of newer switches

✅ **Answer: B** — Old switches default to **dynamic desirable**; newer switches default to **dynamic auto**. Auto + desirable = trunk.

Q2. SW1 is connected to SW2, and SW2 is connected to SW3. You want SW2 to forward SW1's VLAN database information to SW3, but you do not want SW2 to synchronize its VLAN database to SW1. Which command should you use on SW2?

- A. `vtp mode transparent`
- B. `vtp transparent mode`
- C. `vlan mode transparent`
- D. `vtp mode client`

✅ **Answer: A** — `vtp mode transparent` forwards VTP advertisements but does not sync its VLAN database to the server and does not advertise its own VLAN database.

Q3. What are two methods to reset a switch's VTP revision number to 0? Choose two.

- A. Change the VTP domain to an unused domain name
- B. Change the switch to VTP server mode
- C. Change the switch to VTP transparent mode
- D. Use the `vtp reset` command

✅ **Answers: A and C** — Changing to an unused domain or changing to transparent mode resets the revision number to 0. `vtp reset` is not a valid method.

🧠 Boson ExSim Drag-and-Drop: DTP Operational Modes

Use the DTP matrix above. Key answers:

- Access + Access = **Access**
- Access + Dynamic Auto = **Access**
- Access + Dynamic Desirable = **Access**
- Access + Trunk = **Misconfig**
- Dynamic Auto + Dynamic Auto = **Access**
- Dynamic Auto + Dynamic Desirable = **Trunk**
- Dynamic Auto + Trunk = **Trunk**
- Dynamic Desirable + Dynamic Desirable = **Trunk**
- Dynamic Desirable + Trunk = **Trunk**
- Trunk + Trunk = **Trunk**
- Trunk + Access = **Misconfig**

✅ Final takeaway: **DTP negotiates access/trunk status; VTP synchronizes VLAN databases. Know their modes, defaults, revision-number behavior, and when to disable/avoid them.**